

FI360 MASTER ARCHITECTURE & DEVELOPMENT FRAMEWORK

Fleet Intelligence 360 (FI360) — Version 1.0

Fleet Associates Africa Ltd (FAA)
Controlled Product Architecture Document
9 August 2026

Document Control

Field	Value
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Version	1.1
Status	Baseline / Master Architecture — Revised with Flowcharts
Product Owner	Fleet Associates Africa Ltd
Product	Fleet Intelligence 360 (FI360)
Date	9 August 2026
Purpose	Controlling architecture and development framework for all FI360 modules
Applies to	All current and future FI360 modules, services, integrations and platform components

1. Executive Summary

Fleet Intelligence 360 (FI360) is conceived as a modular fleet and asset intelligence ecosystem rather than a single monolithic application. Each FI360 module shall be independently deployable, independently usable, independently licensable and independently maintainable, while being designed from inception to integrate with other FI360 modules through common identity, data standards, APIs and event-driven interfaces.

The commercial principle is: BUY ONE → USE ONE → CONNECT MANY → BECOME FI360. A customer may begin with a single module such as FI360 Tyre Management and later acquire FI360 Workshop & Maintenance without replacing the original module or losing its historical data.

This document establishes the master architectural rules, common services, integration principles, data ownership model, security approach, deployment model, development standards and product roadmap that all FI360 modules must follow.

2. Product Vision and Objectives

2.1 Vision

To create a scalable, intelligent and interoperable fleet-management ecosystem that allows organisations to digitise one operational area at a time and progressively connect their fleet, maintenance, tyres, fuel, parts, people, telematics, compliance and financial intelligence.

2.2 Strategic Objectives

- Provide commercially viable standalone fleet-management modules.
- Protect customer investment when additional modules are purchased.

- Create a common FI360 identity and asset-reference framework.
- Enable secure module-to-module communication through APIs and events.
- Provide consolidated intelligence when multiple modules are connected.
- Support cloud, hosted and appropriate on-premise deployment models.
- Enable configuration for different fleet sizes, industries, branches and operating models.
- Create a reusable development standard that reduces cost and inconsistency across modules.
- Provide a foundation for predictive analytics and AI-assisted fleet decision support.

3. Core Product Principles

Principle	Requirement
Standalone First	Every module must provide useful business value without requiring the full FI360 suite.
Integration Ready	Every module must expose documented interfaces for approved FI360 integrations.
Common Identity	Users, organisations, locations and core assets must have stable FI360 identifiers.
Data Ownership	Each module owns its specialist data; other modules consume approved views/events rather than modifying another module's internal records.
API First	Inter-module and external integration must use controlled APIs/events rather than direct database coupling.
Security by Design	Authentication, authorization, audit and data protection are architectural requirements, not later additions.
Upgrade Independence	A module upgrade should not require simultaneous redevelopment of unrelated modules.
Configuration over Custom Code	Customer-specific variations should be handled through configuration where practical.
Auditability	Material transactions and changes must be traceable to user, time, action and source.
Scalability	The architecture must support growth from small fleets to enterprise and multi-site fleets.

4. FI360 Product Ecosystem

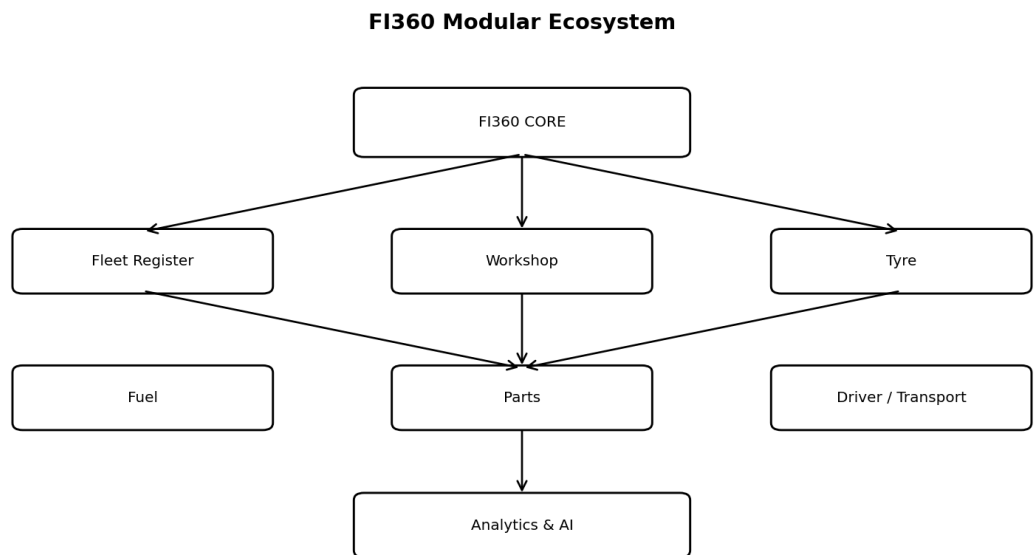


Figure 1. FI360 Modular Ecosystem

Module	Primary Scope	Typical Integrations
FI360 Fleet Register & Asset Management	Asset master, lifecycle, registration and ownership data	All modules
FI360 Workshop & Maintenance	Job cards, work orders, inspections, preventive/corrective maintenance	Fleet, Tyre, Parts, Cost
FI360 Tyre Management	Tyre lifecycle, fitting, removal, inspection, performance and cost	Fleet, Workshop, Fuel, Cost
FI360 Fuel Management	Fuel issues, consumption, reconciliation and fuel analytics	Fleet, Driver, Telematics, Cost
FI360 Parts & Inventory	Stores, parts, issues, receipts, reorder and stock control	Workshop, Fleet, Cost
FI360 Driver Management	Driver records, allocation, compliance and performance	Fleet, Transport, HSE, Telematics
FI360 Transport Operations	Trip planning, dispatch, utilisation and operational control	Fleet, Driver, Fuel, Telematics
FI360 Telematics	GPS, odometer, engine and operational data integration	Fleet, Driver, Fuel, Analytics
FI360 HSE & Compliance	Inspections, incidents, permits, compliance and corrective actions	Fleet, Driver, Workshop
FI360 Cost & TCO	Cost capture, operating cost, lifecycle cost and TCO	All transactional modules
FI360 Fleet Analytics	Cross-module dashboards, KPIs, trends and decision support	All modules
FI360 AI Fleet Advisor	Recommendations, anomaly detection and predictive decision support	Analytics + approved module data

5. Reference Architecture

FI360 shall use a layered architecture. Modules remain bounded business capabilities. The FI360 Core provides shared platform services, while the Integration Layer provides controlled communication between modules and approved external systems.

Reference layers:

- Experience Layer — web/mobile interfaces, dashboards and user interaction.
- Module Layer — independently deployable business modules.
- FI360 Core Layer — identity, organisation, configuration, workflow, notifications, audit and common platform services.
- Integration Layer — API gateway, service interfaces, event bus and integration monitoring.
- Data Layer — module-owned transactional stores plus controlled analytics/data integration stores.
- Intelligence Layer — reporting, BI, analytics, rules, predictive services and AI.
- External Integration Layer — telematics, payment, ERP, accounting, banking, government and other approved systems.

6. FI360 Core

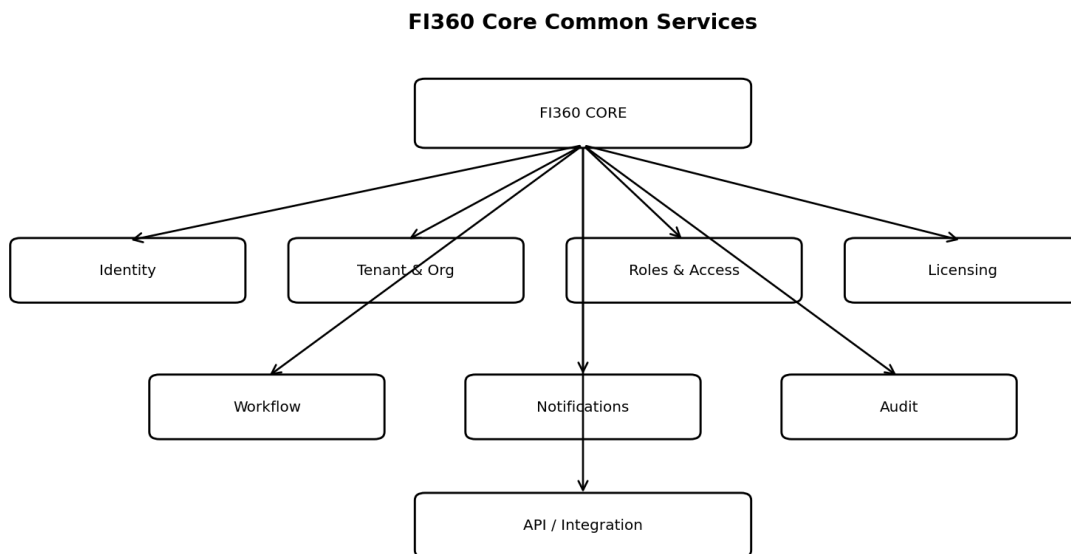


Figure 3. FI360 Core Common Services

6.1 Core Responsibilities

- Tenant and organisation management.
- User identity and authentication.
- Roles and permissions.
- Locations, branches, departments and operational structures.
- Common reference data and configuration.
- Module registration and activation.
- Subscription and entitlement controls.
- Notifications and communication services.
- Workflow and approval framework.
- Audit logging.
- Document and attachment references.
- API credentials and integration governance.
- Platform health and integration monitoring.

6.2 Core Must Not Become a Business Module

The Core shall avoid absorbing specialist business processes that belong to individual modules. For example, tyre inspection logic belongs to Tyre Management and work-order execution belongs to Workshop. The Core provides shared services and references, not duplicated specialist functionality.

7. Standalone Module Architecture

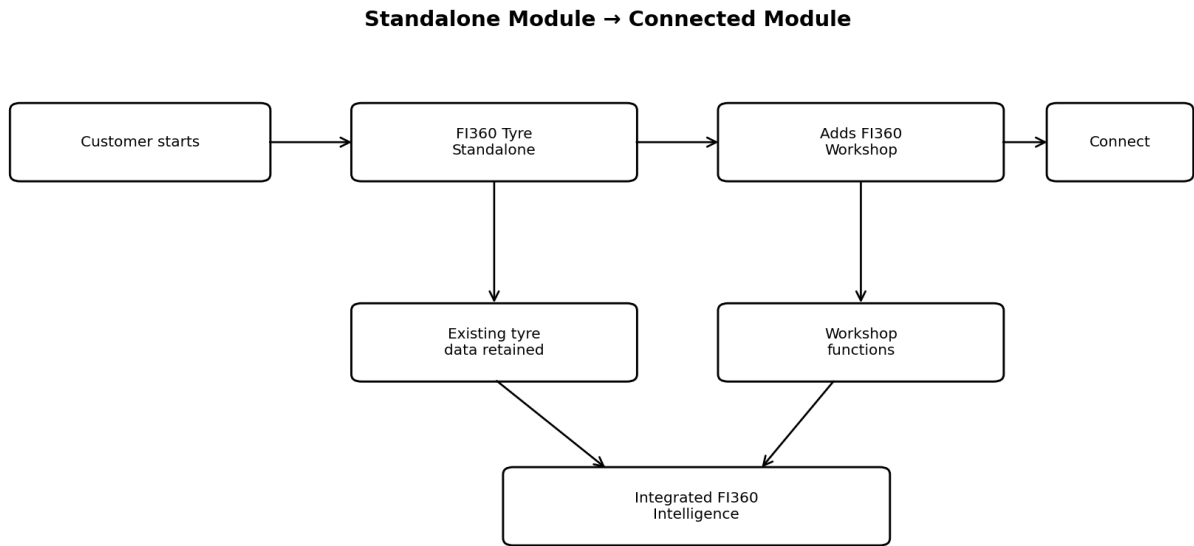


Figure 2. Standalone Module to Connected Module

Each module shall contain its own user-facing functions, business rules, data structures, reports and operational workflows. A module must be capable of being sold and deployed as an independent product.

Component	Standalone Requirement
Application	Own module application/service boundary
Database	Own transactional data store or logically isolated schema
Users	Uses FI360 identity; module-specific permissions may be applied
Dashboard	Module-specific operational dashboard
Reports	Module-specific operational reports
API	Documented APIs for approved integrations
Events	Publishes relevant business events
Audit	Records module transactions and changes
Configuration	Supports module-level configuration
Licensing	Can be independently activated/deactivated

8. Common Identity and Asset Model

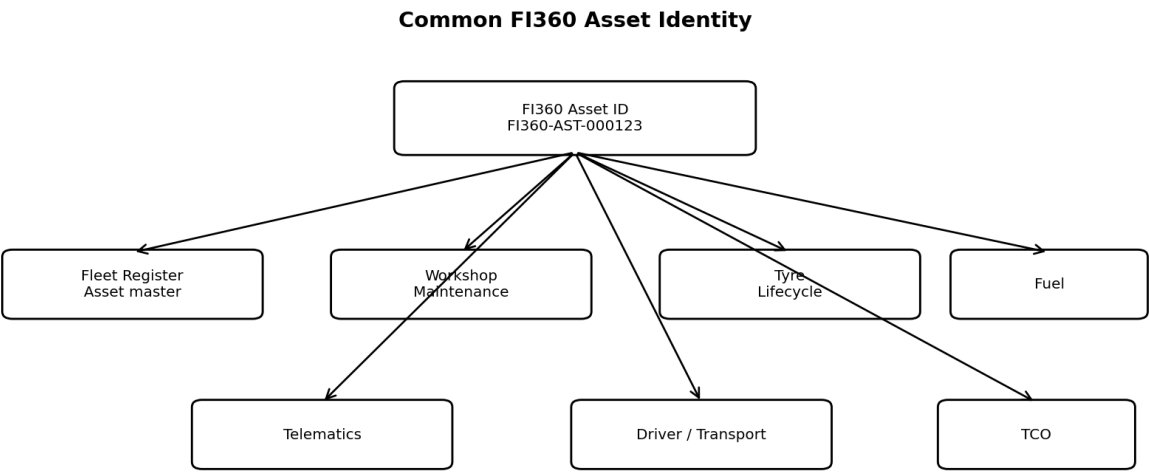


Figure 4. Common FI360 Asset Identity

8.1 Tenant / Customer Identity

Every customer organisation shall receive a stable FI360 Tenant/Organisation identifier. All module records must be traceable to the correct tenant.

8.2 Asset Identity

A stable FI360 Asset ID shall provide the common cross-module reference for vehicles, equipment and other managed assets. Specialist modules may store additional local references, but cross-module transactions must use the canonical FI360 Asset ID.

Core Entity	Example Identifier	Purpose
Tenant	FI360-TEN-000001	Customer organisation
Site	FI360-SIT-000012	Yard, depot, workshop or operating site
Asset	FI360-AST-000123	Vehicle/equipment identity
User	FI360-USR-000045	System user
Driver	FI360-DRV-000078	Driver identity
Tyre	FI360-TYR-000456	Individual tyre identity where serialised
Work Order	FI360-WO-000789	Workshop transaction
Integration	FI360-INT-000015	Configured system-to-system connection

9. Data Ownership and Interoperability

Data Ownership and Controlled Sharing

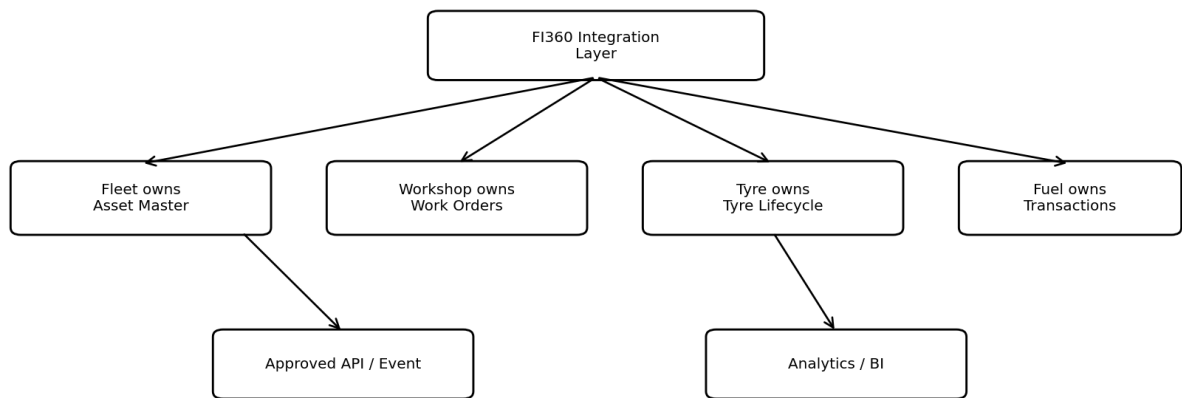


Figure 6. Data Ownership and Controlled Sharing

FI360 shall follow bounded data ownership. A module is the authoritative owner of specialist records within its domain. Other modules access approved data through APIs, shared reference services, events or analytics pipelines.

Domain	Authoritative Module	Examples
Asset master	Fleet Register	VIN, registration, make, model, asset class
Tyre lifecycle	Tyre	Fitting, removal, tread, tyre history
Maintenance execution	Workshop	Work orders, labour, maintenance status
Stock	Parts & Inventory	Receipts, balances, issues
Fuel transactions	Fuel	Fuel issues, litres, fuel cost
Driver profile	Driver	Driver identity, licence and allocation
Trip operations	Transport	Dispatch, trip and operational status
Telematics signals	Telematics	GPS, engine and device data
Cost aggregation	Cost & TCO	Cost structures and TCO calculations

10. API and Integration Architecture

Tyre ↔ Workshop Integration Flow

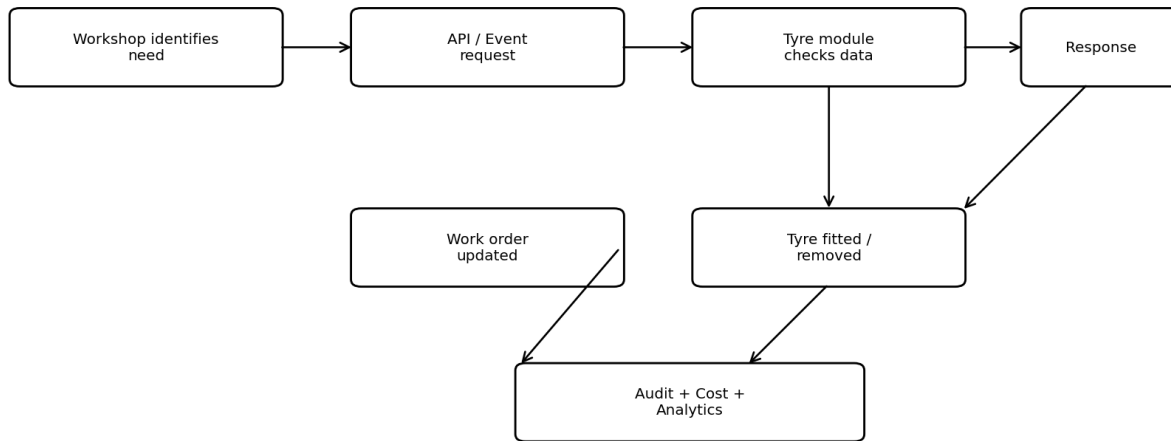


Figure 5. Tyre ↔ Workshop Integration Flow

10.1 API Gateway

All external and inter-module APIs should pass through a controlled API gateway or equivalent integration boundary. Direct production database-to-database dependencies between modules are prohibited.

10.2 API Standards

- Versioned interfaces.
- Authentication and authorization on every protected endpoint.
- Tenant isolation.
- Input validation.
- Consistent error handling.
- Idempotency for appropriate transactional operations.
- Request and correlation identifiers.
- Rate limiting and abuse protection.
- Audit logging for material operations.
- Backward-compatibility rules for supported versions.

10.3 Example Tyre–Workshop Integration

1. Workshop identifies a tyre-related maintenance requirement.
2. Workshop publishes or sends a controlled tyre-replacement request.
3. Tyre module validates the asset and tyre requirements.
4. Tyre module returns availability or a controlled response.
5. Tyre transaction is completed in the Tyre module.
6. Tyre module publishes the fitting/removal event.
7. Workshop associates the event with the relevant work order.
8. Cost/TCO and Analytics receive approved transactional information.

11. Event-Driven Integration

FI360 Event Flow

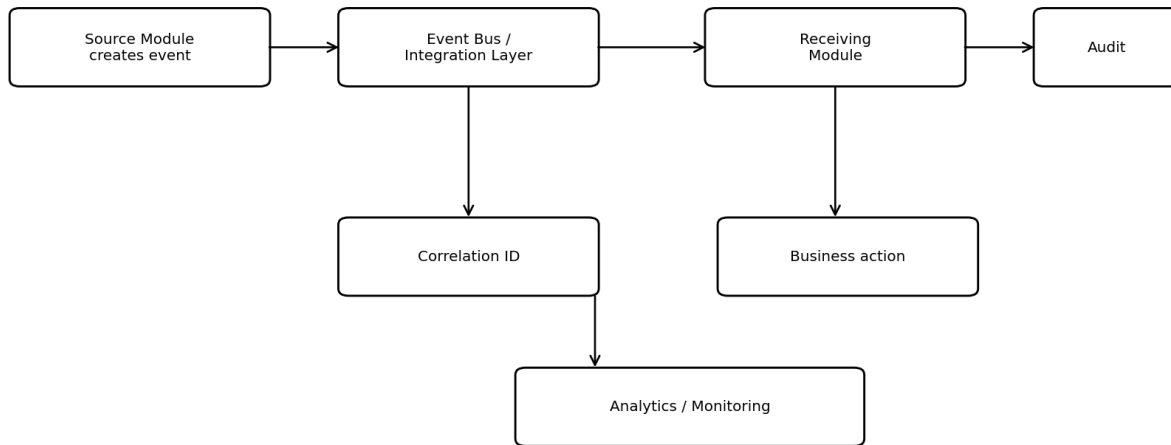


Figure 7. FI360 Event Flow

Where appropriate, modules shall publish business events rather than forcing synchronous calls for every interaction. Events should represent meaningful business facts and should be traceable.

- ASSET_CREATED
- ASSET_UPDATED
- WORK_ORDER_CREATED
- WORK_ORDER_COMPLETED
- TYRE_FITTED
- TYRE_REMOVED
- TYRE_INSPECTION_COMPLETED
- PART_ISSUED
- FUEL_TRANSACTION_POSTED
- DRIVER_ASSIGNED
- TRIP_DISPATCHED
- TELEMATICS_ODOMETER_UPDATED
- COMPLIANCE_ITEM_DUE

The final event catalogue will be controlled in the individual module API and Integration Contracts.

12. Security Architecture

- Role-based access control (RBAC).
- Least-privilege access.
- Tenant isolation.
- Strong authentication and secure session management.
- Encryption in transit and appropriate encryption at rest.
- Secrets and credentials managed outside source code.
- Audit logging of security-sensitive actions.
- Administrative access controls.
- Secure API authentication.

- Regular vulnerability management and dependency updates.
- Backup, recovery and business-continuity controls.

Security requirements shall be refined during the detailed SRS and technical design stages.

13. Workflow and Approval Framework

FI360 shall support configurable workflow states and approvals where operational or financial control requires them.

Workflow Area	Example States
Workshop	Draft → Submitted → Approved → In Progress → Inspection → Completed → Closed
Tyre	Requested → Approved → Issued → Fitted → Active → Removed → Retired
Parts	Requested → Approved → Issued → Returned/Consumed
Fuel	Captured → Validated → Reconciled → Posted
Compliance	Open → Assigned → Evidence Submitted → Reviewed → Closed

14. Reporting and Intelligence

Each module must provide operational reporting independently. When modules are connected, FI360 Analytics shall provide cross-module intelligence without changing the source-of-record ownership of transactional modules.

- Operational dashboards per module.
- Management KPI dashboards.
- Exception reporting.
- Trend analysis.
- Cost analysis.
- Asset lifecycle analysis.
- Utilisation and availability analysis.
- Predictive and prescriptive analytics in later phases.

15. Audit, Logging and Traceability

Every material transaction shall be traceable through an audit record containing, where applicable:

- Tenant/organisation.
- User or system identity.
- Timestamp.
- Module.
- Transaction/entity identifier.
- Action performed.
- Previous and new status/value where relevant.
- Source channel or integration.
- Correlation/reference ID.

16. Multi-Tenant and Organisation Model

FI360 should be designed for multi-tenant SaaS operation while allowing suitable dedicated deployments where commercially or technically required.

- Tenant isolation is mandatory.
- A tenant may have multiple sites, branches, departments and fleets.
- Users may have access to one or multiple sites according to permissions.
- Enterprise customers may have multiple legal or operating entities.
- Reporting access must respect organisational boundaries.

17. Deployment Architecture

Model	Use Case
Cloud SaaS	Default model for scalable subscription deployments.
Dedicated Cloud	Enterprise customers requiring dedicated environments.
Hybrid	Customers integrating FI360 with existing on-premise systems.
On-Premise / Private Deployment	Only where justified by customer, regulatory or infrastructure requirements.

The product architecture should avoid unnecessary dependence on a single deployment model.

18. Module Lifecycle and Versioning

9. Concept and business case.
10. Module architecture.
11. Business process design.
12. SRS and functional requirements.
13. Data model and API contract.
14. UI/UX specification.
15. Development.
16. Unit and integration testing.
17. Security testing.
18. User acceptance testing.
19. Release and deployment.
20. Monitoring and support.
21. Versioned enhancement and maintenance.

Every module must maintain compatibility rules for its published APIs. Breaking changes require a new API version and a documented migration/deprecation path.

19. Module Development Standard

FI360 Module Development Lifecycle

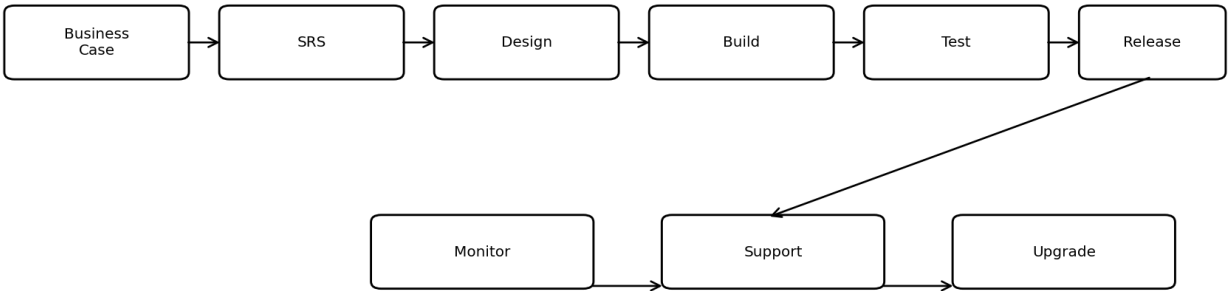


Figure 9. FI360 Module Development Lifecycle

Every FI360 module shall have a controlled development package containing:

- Module Business Case.
- Module Scope and Objectives.
- Business Process Map.
- Stakeholder and User Roles.
- Functional Requirements.
- Non-Functional Requirements.
- Use Cases/User Stories.
- Data Model.
- Data Dictionary.
- UI/UX Specification.
- Workflow Specification.
- Reporting and KPI Specification.
- Security Specification.
- API Specification.
- Integration Contract.
- Event Catalogue.
- Test Strategy and Test Cases.
- Developer Backlog.
- Deployment and Support Guide.

20. Testing Framework

Test Level	Purpose
Unit	Validate individual functions/components.
Module Integration	Validate internal module interactions.
API Contract	Validate published interfaces and compatibility.
Cross-Module	Validate FI360 module interoperability.
Security	Validate authentication, authorization and common

	security controls.
Performance	Validate response times, throughput and scalability.
User Acceptance	Validate operational requirements with representative users.
Regression	Ensure new releases do not break supported functions/integrations.

21. Observability and Support

- Application health monitoring.
- API availability and latency monitoring.
- Integration failure monitoring.
- Job/event processing monitoring.
- Error logging with correlation IDs.
- Operational alerts.
- Backup monitoring.
- Usage and module adoption metrics.

22. Commercial and Subscription Architecture

FI360 shall support modular licensing. A customer subscription may activate one or more modules. Subscription entitlements must be centrally controlled while module business functionality remains independently deployable.

Commercial Package	Example
Single Module	FI360 Tyre only
Module Bundle	Tyre + Workshop + Parts
Enterprise Suite	Multiple or all FI360 modules
Add-On	Analytics, Telematics or AI services
Integration Service	External system/telematics integration

23. Customer Data Protection and Exit

Customer data portability must be considered from the beginning. A customer should be able to export agreed data in documented formats subject to contractual and legal requirements. Module integration must not create unnecessary lock-in through undocumented proprietary data structures.

24. AI and Analytics Architecture

AI services shall consume governed and approved data from FI360 modules and analytics stores. AI outputs must be distinguishable from source transactional records and should provide traceability to the underlying data and applicable business rules.

- Predictive maintenance.
- Tyre replacement prediction.
- Fuel anomaly detection.
- Vehicle downtime prediction.
- Maintenance cost forecasting.
- Fleet utilisation optimisation.
- Exception prioritisation.
- Management recommendations.

25. Future External Integrations

The architecture shall be capable of integrating with approved external systems such as telematics platforms, ERP/accounting systems, payment services, mapping services and relevant government or regulatory systems. Each integration shall have its own controlled connector/API contract.

Potential integration categories include GPS/telematics, accounting/ERP, fuel-card systems, mobile money/payment services, identity services, mapping and regulatory data sources.

26. FI360 Product Roadmap

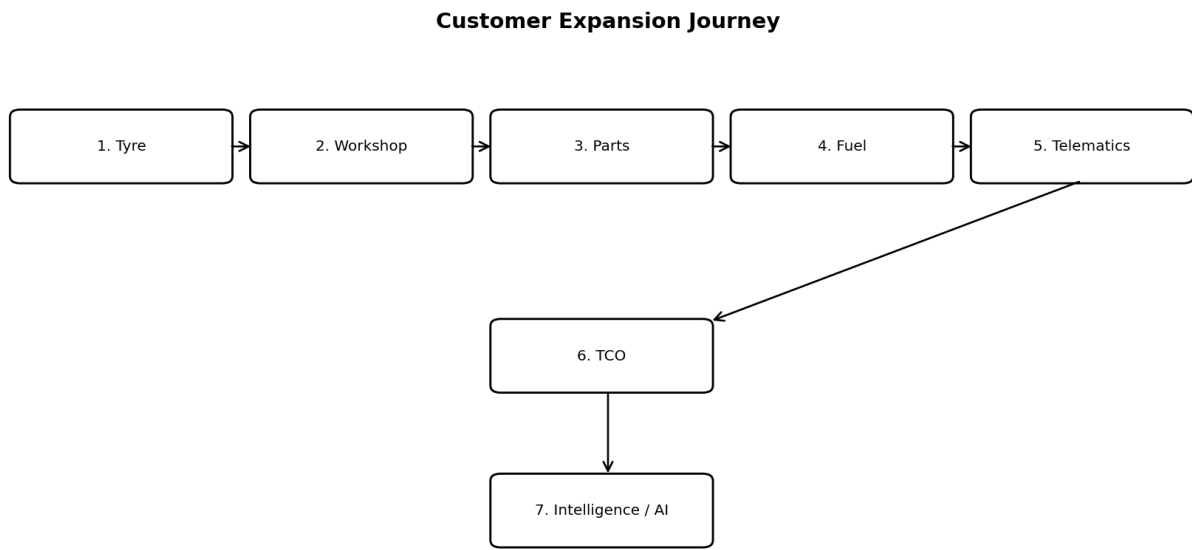


Figure 8. Customer Expansion Journey

Phase	Primary Deliverables
Phase 1	Master Architecture, FI360 Core standards, common data and API standards
Phase 2	FI360 Tyre Management standalone MVP + API/Integration Contract
Phase 3	FI360 Workshop & Maintenance + Tyre/Workshop integration
Phase 4	Fleet Register + Parts + Fuel integration
Phase 5	Driver + Transport + Telematics
Phase 6	Cost/TCO + Compliance
Phase 7	Fleet Analytics and executive intelligence
Phase 8	AI Fleet Advisor and predictive capabilities

27. Governance and Change Control

This Master Architecture is the controlling reference for FI360 product development. Any proposed architectural exception must be documented, justified, reviewed and approved by the FI360 product/technical governance function.

Governance Item	Requirement
Architecture change	Document rationale and impact
API change	Version and document compatibility impact
Data model change	Assess affected modules and migrations

Security change	Security review before release
New module	Must comply with Module Development Standard
External integration	Must have approved connector/API contract

28. Key Architectural Decisions

22. FI360 will be modular rather than monolithic.
23. Every module must work independently.
24. Every module must be integration-ready from inception.
25. FI360 Core provides common platform services.
26. Modules retain ownership of their specialist transactional data.
27. Stable FI360 identifiers provide cross-module references.
28. Inter-module communication will use controlled APIs and/or events.
29. Direct production database coupling between modules is prohibited.
30. Module licensing can be independent.
31. Customers can progressively expand their FI360 footprint.
32. Analytics and AI will consume governed data rather than becoming the source of transactional truth.

29. Definition of Done — Architecture Level

The FI360 Master Architecture will be considered ready for module development when:

- The modular principles are approved.
- The FI360 Core responsibilities are defined.
- The common identity model is defined.
- Asset identity and cross-module references are defined.
- API and event principles are established.
- Security and tenant-isolation principles are established.
- The module development template is approved.
- The first module is selected and scoped.
- The architecture supports standalone and connected deployment.

30. Immediate Next Development Step

The next controlled deliverable after this Master Architecture is DOCUMENT 02 — FI360 CORE ARCHITECTURE & COMMON SERVICES SPECIFICATION. It will define the technical foundation shared by every module, including tenant management, identity, users, roles, permissions, module registry, subscription/entitlement management, common asset references, workflow, notifications, audit, API gateway and integration governance.

Following Document 02, development will proceed to DOCUMENT 03 — FI360 MODULE DEVELOPMENT STANDARD, followed by the complete FI360 Tyre Management module package.

Appendix A — Initial FI360 Integration Example

Scenario: Customer starts with FI360 Tyre Management and later adds FI360 Workshop & Maintenance.

Step	Tyre Module	Workshop Module	FI360 Platform
1	Customer manages tyres independently	Not installed	Manages tenant/user identity
2	Stores tyre and asset references	Installed later	Provides module entitlement

3	Publishes tyre events	Consumes approved events	Routes/monitors integration
4	Records tyre fitting/removal	Associates event with work order	Maintains audit/correlation
5	Maintains tyre cost/history	Records maintenance activity	Feeds analytics when enabled
6	Continues independent operation	Continues independent operation	Presents integrated intelligence

Appendix B — Architectural Principle Statement

“Every FI360 module must be useful alone, but designed to become more valuable when connected.”

Document End

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Revision 1.1 — Flowchart Enhancement

Version 1.1 retains the approved Version 1.0 architecture and adds embedded visual flowcharts at key architecture points. The diagrams clarify the modular ecosystem, standalone-to-connected model, FI360 Core, common Asset ID, data ownership, API/event integration, customer expansion and module development lifecycle.

These diagrams are explanatory architecture views; detailed technical sequence diagrams, ERDs and deployment diagrams will be produced in the relevant Core and module technical specifications.